

Assignment 3 Responses:

- Naive_mae produced a validation of 10.3439
- Dense_Mae = .7920747
- LSTM_mae = 1.055111
- GRU-mae = .8321033
- Stacked_mae = .601744
- CONV_mae = .5960037

Takeaways from above:

- Dense model performed better than LSTM and GRU
- LSTM was the worst performing model
- Stacked GRU and Conv1D+GRU both performed about the same and were the best performing overall of the models

Model Units under GRU: 16, 32, 64, 128

- Models performed better from 16->32->64 but then got worse at 128

```
gru_unit_results
```

```
## # A tibble: 4 x 2
##   Model      MAE
##   <chr>      <dbl>
## 1 GRU (16 units) 1.97
## 2 GRU (32 units) 1.48
```

```
## 3 GRU (64 units) 0.917
## 4 GRU (128 units) 1.03
```

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Model Units under Conv1D+GRU: 16, 32, 64, 128

- Each model got better over time, potentially could have gotten even better with one more increase but did not investigate further

```
conv_gru_unit_results
```

```
## # A tibble: 4 x 2
##   Model                MAE
##   <chr>              <dbl>
## 1 Conv1D+GRU (16 units) 1.19
## 2 Conv1D+GRU (32 units) 1.000
## 3 Conv1D+GRU (64 units) 0.640
## 4 Conv1D+GRU (128 units) 0.405
```

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- Best Model Overall was Conv1D+GRU at 128 units =.405

```
##
## Test set evaluation for: Conv1D+GRU (128 units)
##   loss      mae
## 0.1945383 0.3259089
```

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Takeaways from all results:

- Neural networks outperformed baseline
- Dense model performed well
- Conv1D+GRU was best overall performance
- Interesting takeaway: At 128 GRU overfits but Conv1D performs better.